

Change the intended text nodes - and prove everything else stayed within plan.

PPTrans turns translation into a source-bound transaction across untrusted OOXML and AI output.

ENGLISH SOURCE

TRANSLATION

Translate PowerPoint.
Preserve the PowerPoint.

PPTrans patches only planned text nodes, verifies the package, and never overwrites the source deck.

PPTrans v2 • unreleased local showcase

01

CURATED ZH-CN

翻译 PowerPoint。
保留 PowerPoint 结构。

PPTrans 只改目标文本，验证 OOXML，绝不覆盖源文稿。

PPTrans v2 • 未发布的本地展示版

01

Real changed-text fixture through the exact-ID patch and verification path. Native LibreOffice 26.8.0.3 renders; this is not provider-quality evidence.



WHAT I ENGINEERED

- Led v2 product direction and the end-to-end re-architecture around immutable **DeckPlan** and **PatchSet** contracts.
- Built defensive ZIP/XML intake, source-hash binding, stable text addresses, and explicit support and resource limits.
- Made OpenAI and Anthropic adapters fail closed on partial, reordered, duplicated, invented, or schema-invalid IDs.
- Added staged patching, planned-and-untouched text verification, package checks, and atomic no-clobber publication.

NO-SPEND PROVIDER PREVIEW

41 units in 2 logical calls | 2,799 source/context chars | 8,867 serialized request chars | largest 5,903

Zero-memory-hit upper bounds. The preview loads no credential, provider SDK, translation memory, or output path and makes no provider/API request. Counts are not tokens, price, latency, availability, or quality estimates.

MEASURED EVIDENCE

541

PASSING TESTS

92.12%

BRANCH-AWARE COVERAGE

9,346

GENERATED EXAMPLES

3 / 41 / 45

SLIDES / UNITS / SPANS

3

CHANGED SLIDE XML PARTS

58.770 / 60.972 ms

MEDIAN / P95 LOCAL CORE

EVIDENCE BOUNDARY

- The synthetic, author-reviewed zh-CN deck proves changed text and package preservation, not production-provider translation quality.
- The benchmark covers the deterministic local core only; it excludes provider, network, memory, rendering, cost, and translation-quality time.
- LibreOffice evidence does not claim Microsoft PowerPoint pixel identity, automatic layout fit, universal compatibility, or maximum practical deck size.

STATUS AND PROVENANCE

PPTrans v2 is an **unreleased, non-public local engineering showcase**. The public main branch is legacy and is intentionally not linked here. Repository history is not presented as clean-room work; another package or release remains blocked pending upstream provenance/licensing clarification documented in NOTICE.md.

Configured Python support: 3.10-3.13 | Skills: OOXML/XML | typed domain design | schema-bound LLM adapters | transactional I/O | property testing | CI and security